

CMPG 325 - Network Design Project

Tlou Furniture & Appliances (Rustenburg) - Client ID: CLI-017

Phase 1

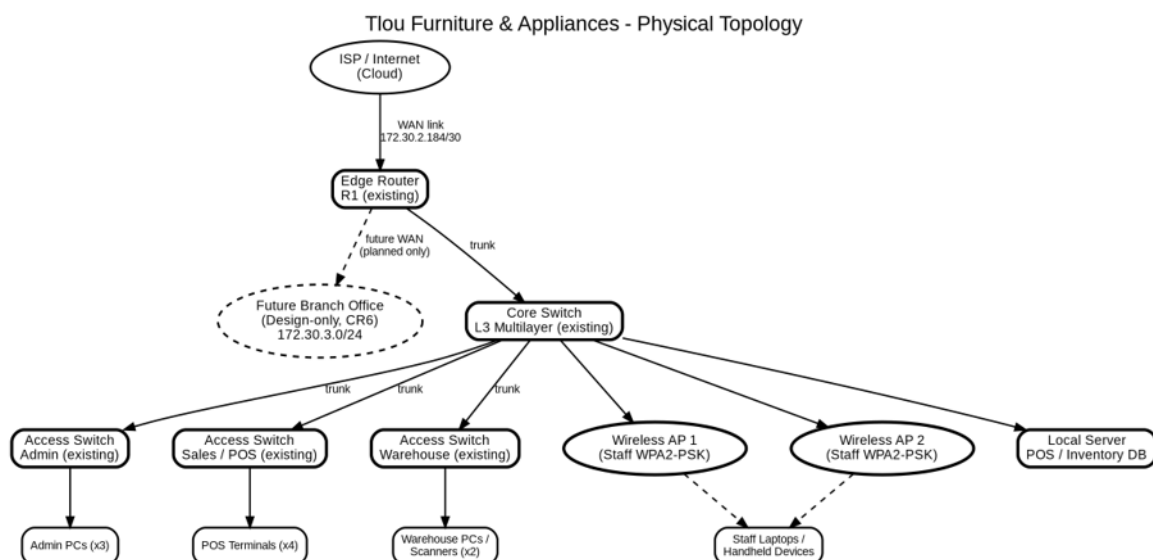
1.Client Requirements

Tlou Furniture & Appliances is a retail organisation based in Rustenburg. The network design must meet the following client requirements as specified in the project brief:

- Assigned organisation: Tlou Furniture & Appliances (Rustenburg), Retail industry.
- Use the assigned addressing block 172.30.2.0/23 as the basis of the IP addressing plan.
- Provide appropriate connectivity and network services for Admin, Sales/POS and Warehouse operations.
- Accommodate a limited equipment budget, existing switches must be reused where possible rather than purchasing new hardware.
- Configure, verify and demonstrate the assigned networking challenge: Wireless Security (WPA2-PSK hardening, with an optional Enterprise-level extension).
- Accommodate Client Change Request CR6: a branch office is planned. The design and addressing plan must show how this is accommodated, but no second-site build is required, design/addressing only.
- Produce a working testable Cisco Packet Tracer implementation with demonstrated end-to-end connectivity

2.Physical Topology

The physical topology reflects the actual devices, cabling, and equipment reuse constraint. The existing core switch and three access switches are retained; only wireless access points are treated as new equipment to meet the WPA2-PSK requirement. The future branch office is shown only as a planned WAN connection point, in line with CR6.



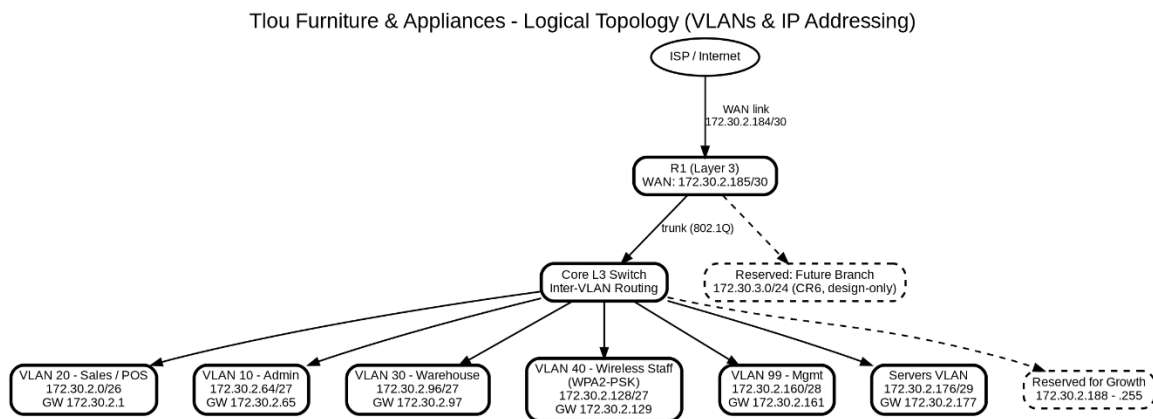
Key design decisions:

- Existing core switch reused as the Layer 3 distribution point to minimise new equipment spend.
- Three existing access switches retained for Admin, Sales/POS and Warehouse - matching the budget constraint.
- Two wireless access points added to support staff wireless connectivity secured with WPA2-PSK.

A single local server hosts POS and inventory database services on-site

3.Logical Topology

The logical topology shows how the network is segmented into VLANs for traffic separation and security, with inter-VLAN routing handled by the core Layer 3 switch. Each department and function is isolated onto its own VLAN and subnet reducing broadcast traffic and containing the wireless segment separately for security hardening.



VLAN Summary

- VLAN 20 - Sales / POS: point-of-sale terminals and retail floor devices.
- VLAN 10 - Admin / Management: office and management staff devices.
- VLAN 30 - Warehouse / Inventory: stock and inventory devices.
- VLAN 40 - Wireless Staff: isolated wireless segment secured with WPA2-PSK
- VLAN 99 - Network Management: dedicated management VLAN for switches and access points.

Servers VLAN: hosts the local POS/inventory database server

4.IP addressing plan

The IP addressing plan uses Variable Length Subnet Masking (VLSM) to divide the assigned 172.30.2.0/23 block efficiently across VLANs, measured to each segment's actual host requirements. Address space is deliberately reserved for future growth and for the CR6 branch office change request.

Subnet / VLAN	Network	Usable Range	Broadcast	Gateway
VLAN 20 - Sales / POS	172.30.2.0/26	.1 - .62	172.30.2.63	172.30.2.1
VLAN 10 - Admin / Management	172.30.2.64/27	.65 - .94	172.30.2.95	172.30.2.65
VLAN 30 - Warehouse / Inventory	172.30.2.96/27	.97 - .126	172.30.2.127	172.30.2.97
VLAN 40 - Wireless Staff (WPA2-PSK)	172.30.2.128/27	.129 - .158	172.30.2.159	172.30.2.129
VLAN 99 - Network Management	172.30.2.160/28	.161 - .174	172.30.2.175	172.30.2.161
Servers VLAN (POS/Inventory DB)	172.30.2.176/29	.177 - .182	172.30.2.183	172.30.2.177
WAN Link (Router - ISP)	172.30.2.184/30	.185 - .186	172.30.2.187	172.30.2.185

Address Space Reserved for Future Use

- 172.30.2.188 – 172.30.2.255: reserved for future growth at the Rustenburg site, for example additional VLANs or devices.
- 172.30.3.0/24: reserved for the future branch office per CR6. No devices are configured against this block.

GitHub repo: <https://github.com/Lathitha27/Computer-Networks>